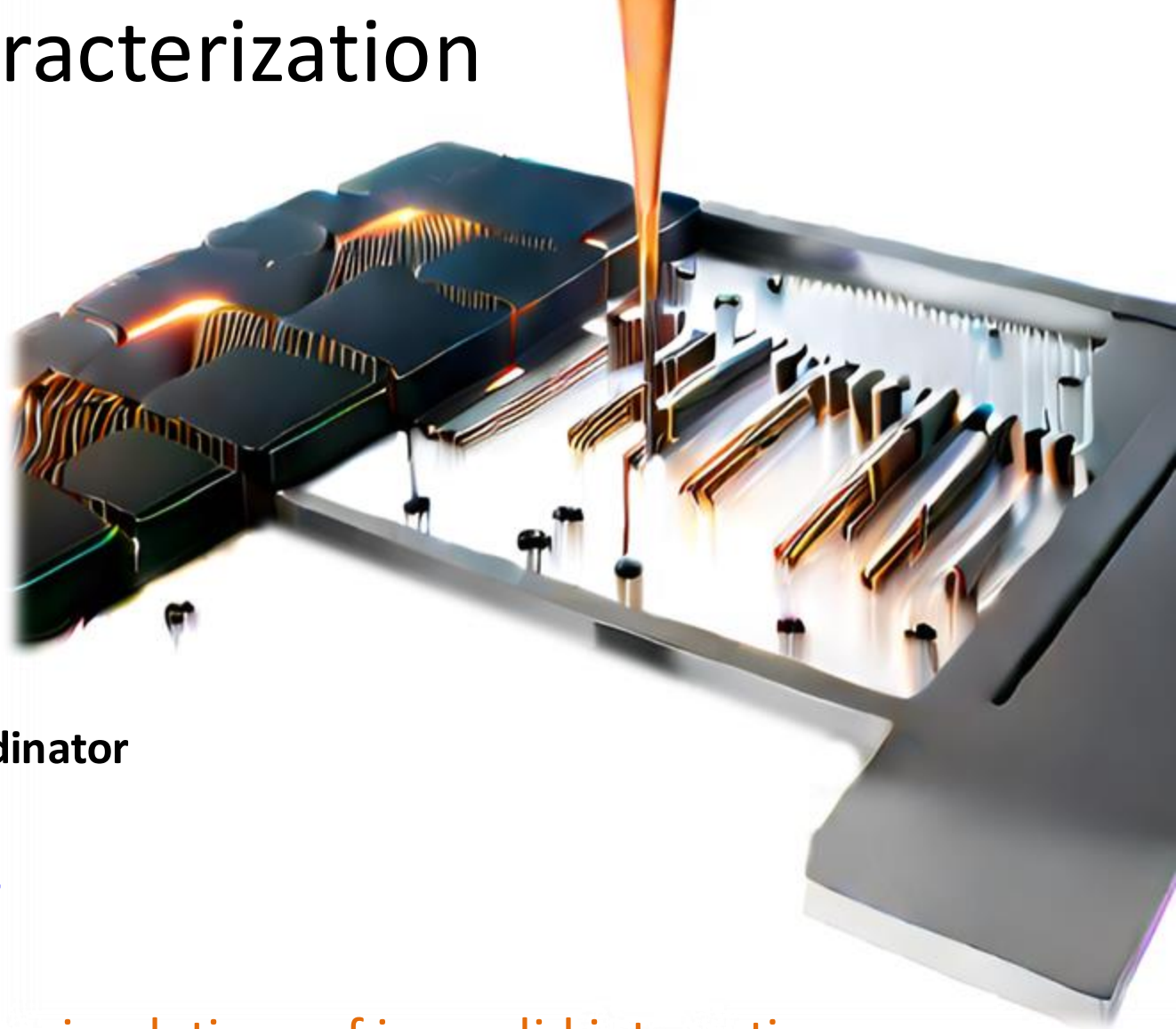


68320 Nanofabrication and and Nanocharacterization Techniques



UNIVERSITY OF
TECHNOLOGY SYDNEY



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Tutorial 3: Monte Carlo simulations of ion-solid interactions

Ions vs electrons

electrons

light ions

heavy ions

mass; momentum: $p = mv$ (at a given energy)

velocity: $E = 0.5 mv^2$ (at a given energy)

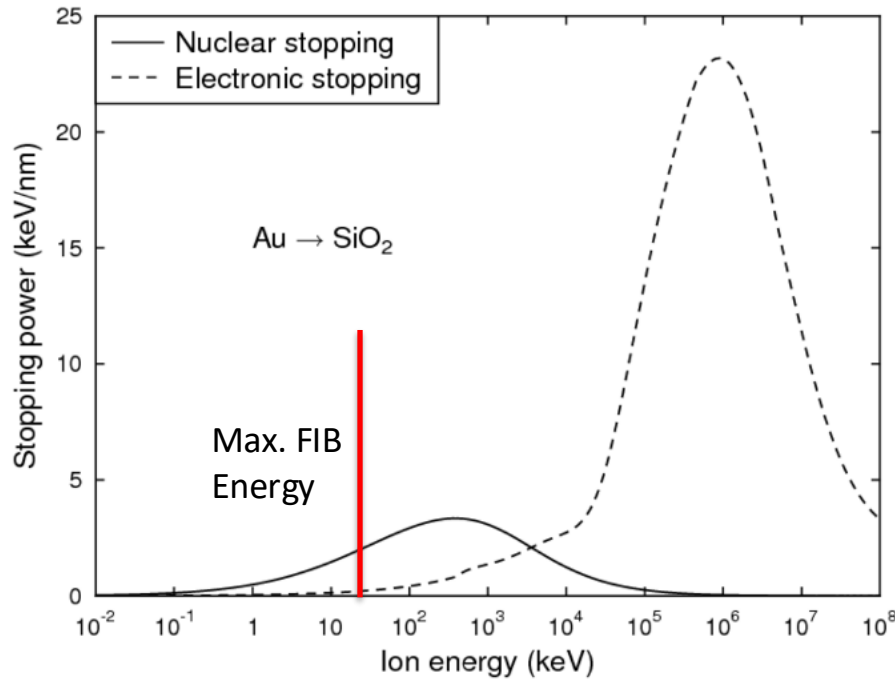
penetration range (in a sample)

sample interactions & DAMAGE

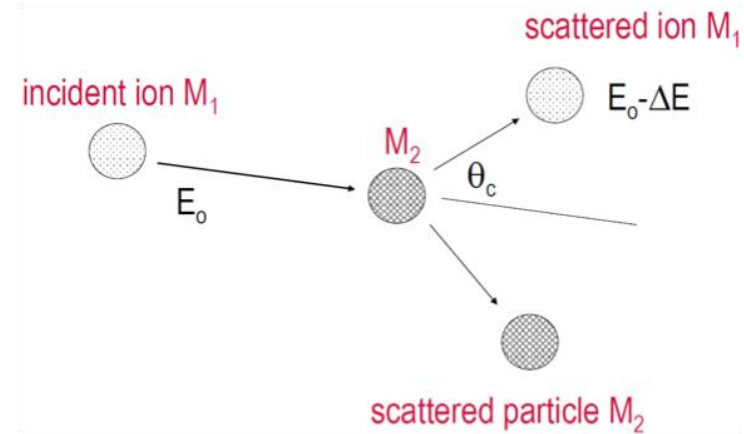
nanofabrication SPEED (throughput)

ions can remain in the sample – ion “implantation” / “doping”

Ion-solid interactions



ion-atom collision (nuclear stopping)



Nuclear stopping

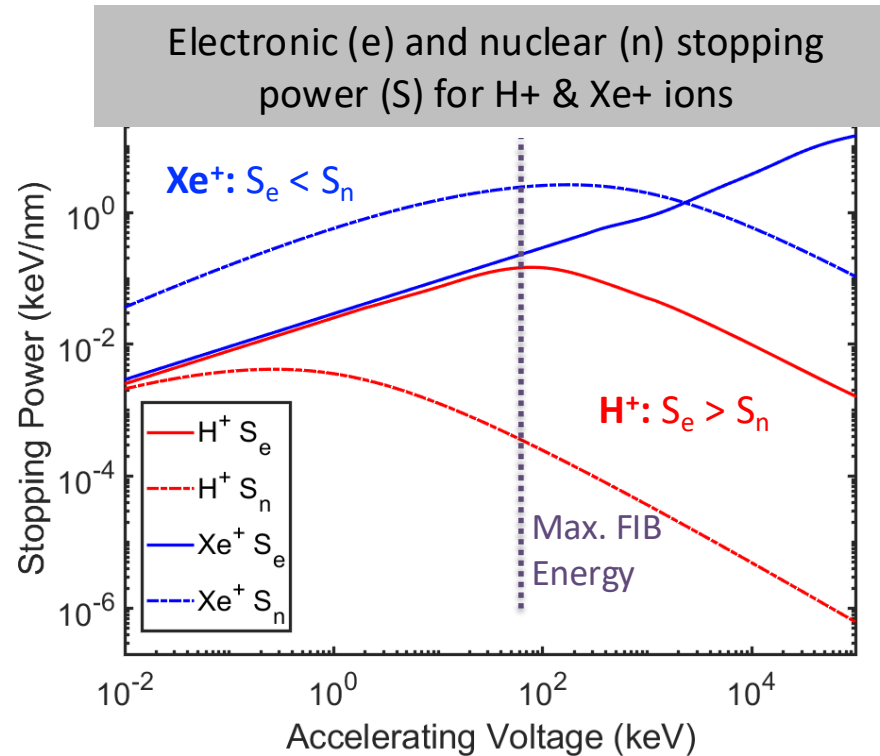
- Causes displacement of atoms and sputtering of the sample.
- Useful for ion beam milling of samples.

Electronic stopping

- Ions lose energy via inelastic interactions with electrons.
- Mainly due to Coulombic forces.
- Causes production of secondary electrons.

(“stopping power” = energy lost per unit distance travelled in a sample)

Ion-solid interactions



Protons

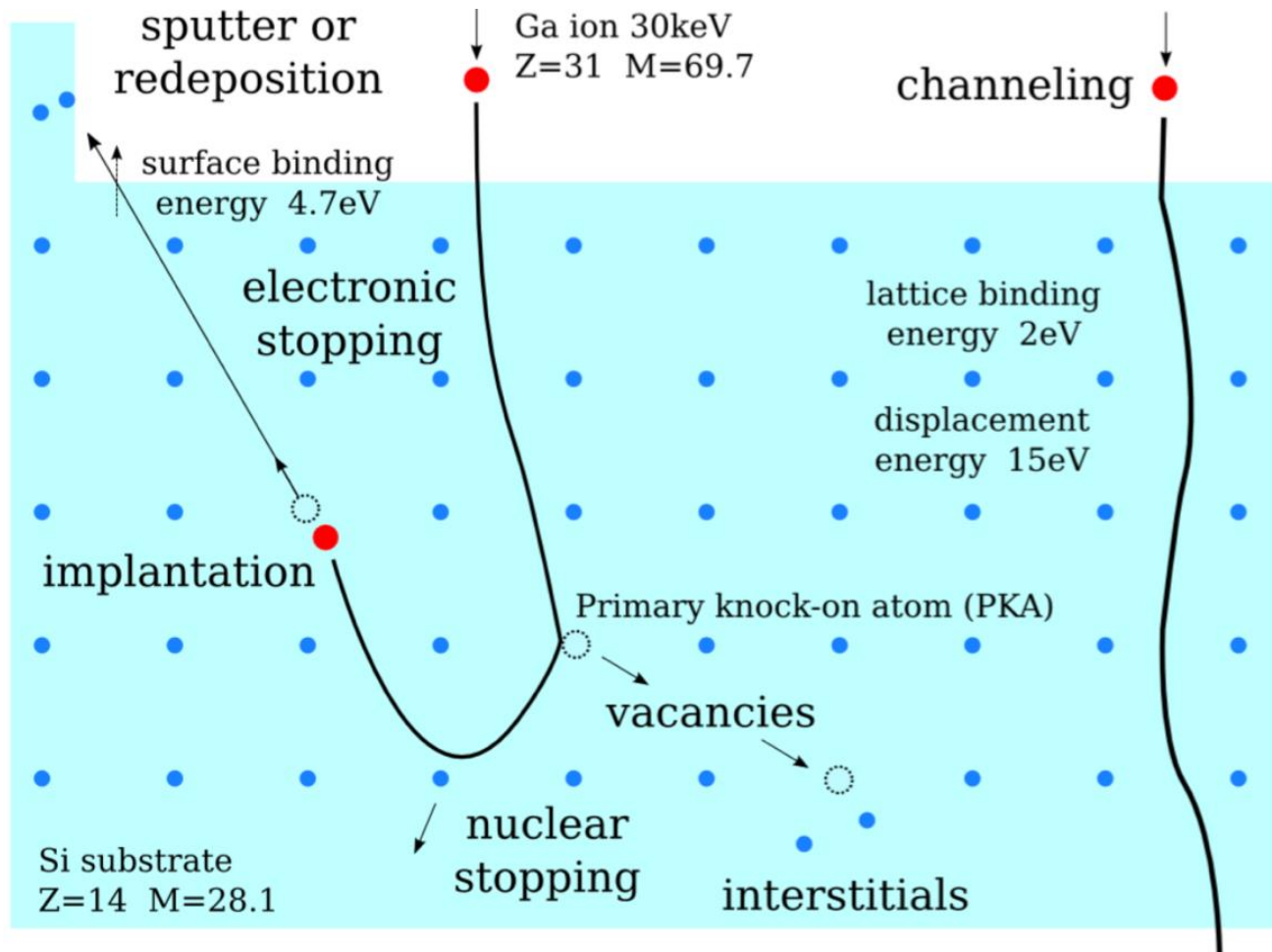
- Electronic scattering dominant at FIB energies (~ 1 to 50 keV)
- Total stopping power low $\rightarrow$ long ion range in samples
- Bad for milling of samples, good for generating secondary electrons and X-Rays

Heavier Ions

- Nuclear scattering dominant at FIB energies (~ 1 to 50 keV)
- Total stopping power high = short range
- Useful for ion beam milling of samples

(“stopping power” = energy lost per unit distance travelled in the solid)

Ion-Solid interactions



Electrons vs Ions



- Small mass
- High speed at a given energy but low momentum
- Long penetration range
- Atomic displacements unlikely

- High mass
- Low speed at a given energy but high momentum
- Short penetration range
- Atomic displacements very likely
- Sputtering/milling
- Ion implantation

Kinetic energy

$$E = q\Delta V$$

Kinetic energy

$$E = 0.5 mv^2$$

Momentum

$$p = mv$$

ΔV = potential difference used to accelerate electron/ion beam (e.g. 10 kV)

$|q|$ = charge (the same for an electron & a singly ionized ion)

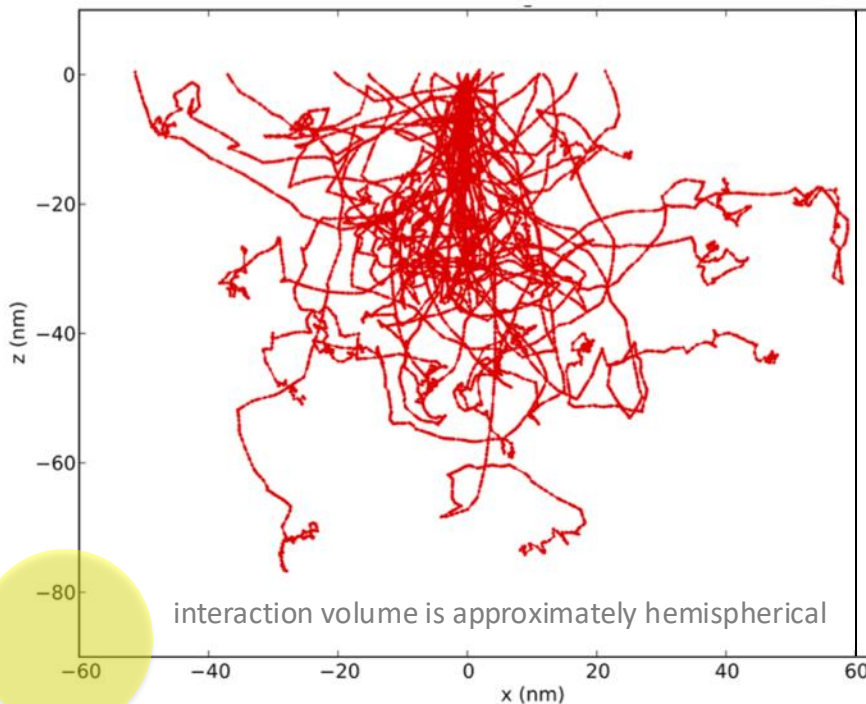
m = mass ($m_{\text{ion}} \gg m_{\text{electron}}$)

Ion-Solid interactions

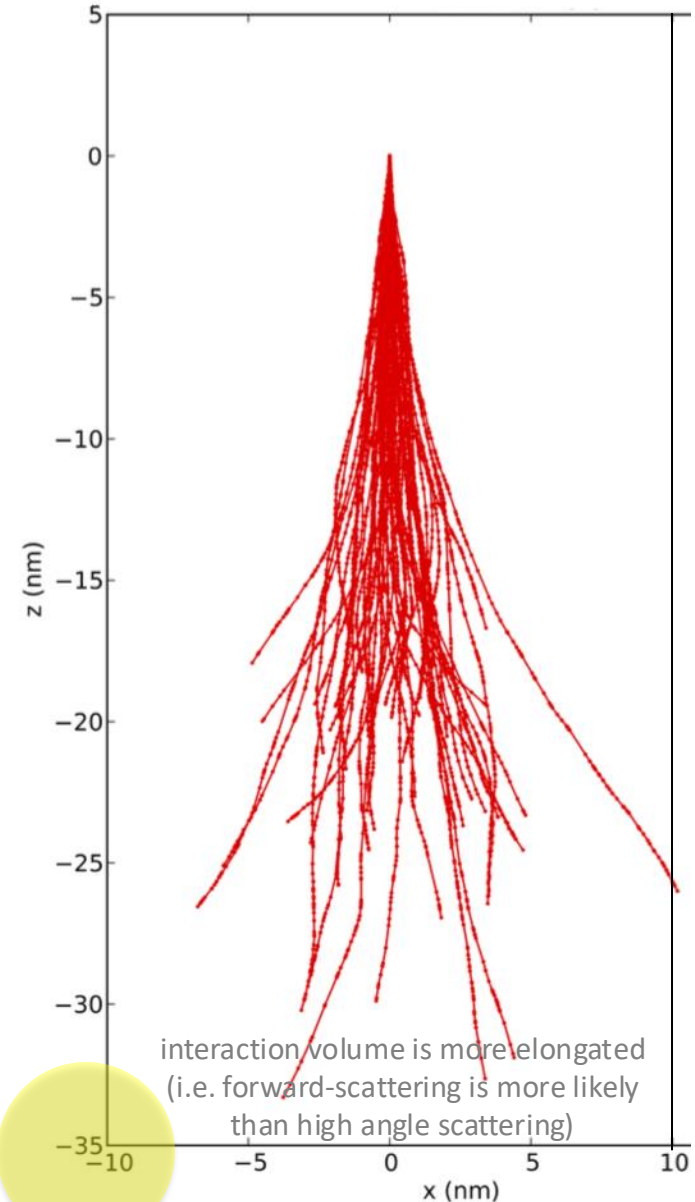
Ion interaction volume depends on ion species & composition of the sample (often called a “target” in the literature).

eg: Carbon ($M=12.0107$) and Tungsten ($M=183.84$)

30 keV C+ onto Tungsten



30 keV W+ onto Carbon

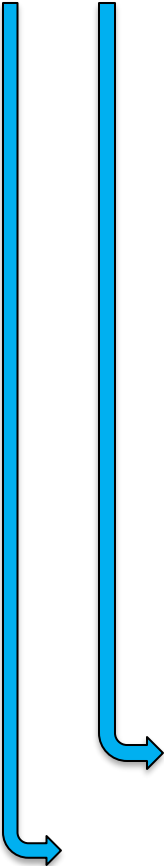


Monte Carlo Simulation of Ion Interactions With Solids

Free software called “SRIM”: <http://www.srim.org>



Extensive documentation & tutorials; SRIM is used widely in research literature



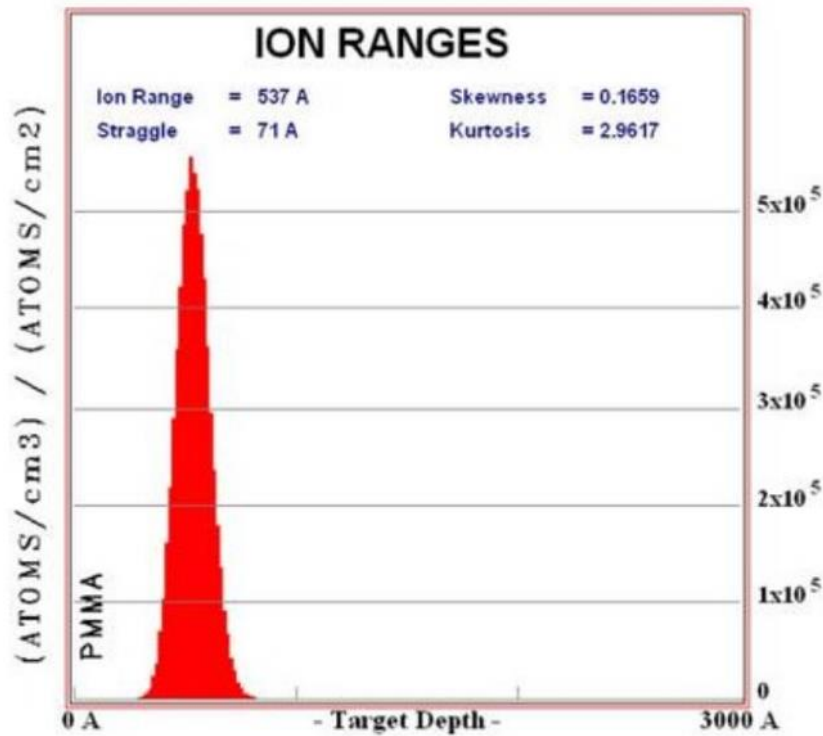
NEW ! 100 Years of Ion Stopping 2300+ Papers Listed and Results Plotted !!	
SRIM Textbook	
↓ Software ↓	↓ Science ↓
SRIM / TRIM Introduction	Historical Review
Download SRIM-2013	Details of SRIM-2013
SRIM Install Problems	Stopping in Compounds
SRIM Tutorials	High Energy Stopping Theory & Experiments
Download TRIM Manual Part-1, Part-2	Ranges of Ions
Stopping, Range and Damage by Neutrons	Including SRIM in Other Software Programs
SRIM - Supporting Analytic Software	SREM - Stopping and Range of Electrons

NOTE: the SRIM GUI is terrible, but the underlying physics model is very good – don't be put off by the awful graphics

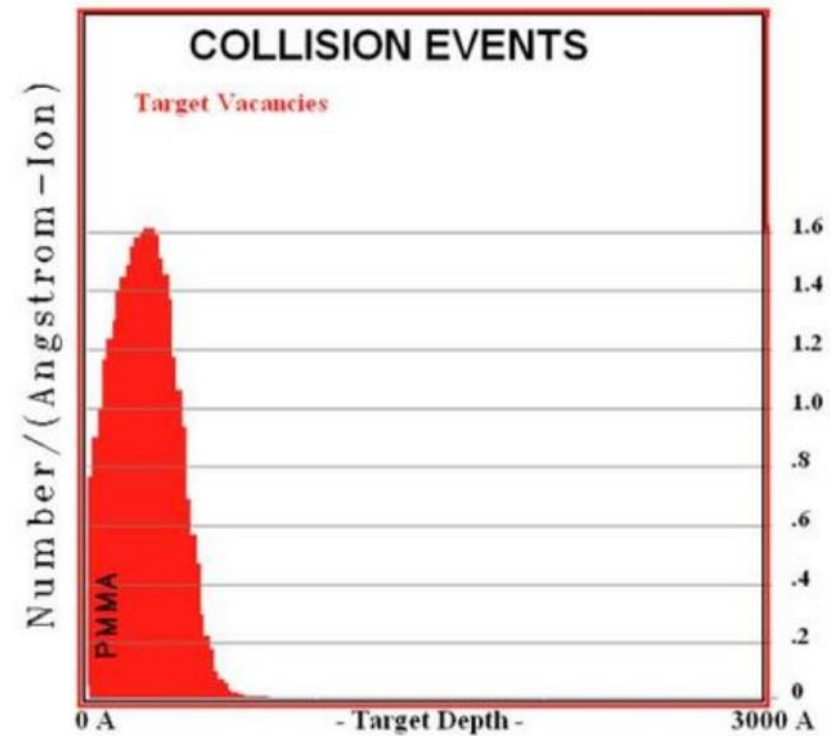
Example of SRIM outputs

PMMA (“polymethyl methacrylate polymer”) bombarded by 40 keV Au⁺ ions

depth distribution of Au atoms implanted in PMMA



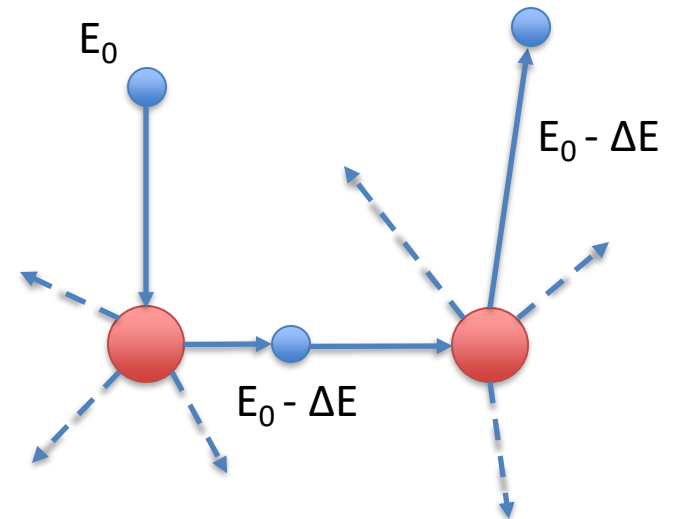
depth distribution of vacancies generated by Au ions



SRIM vs CASINO

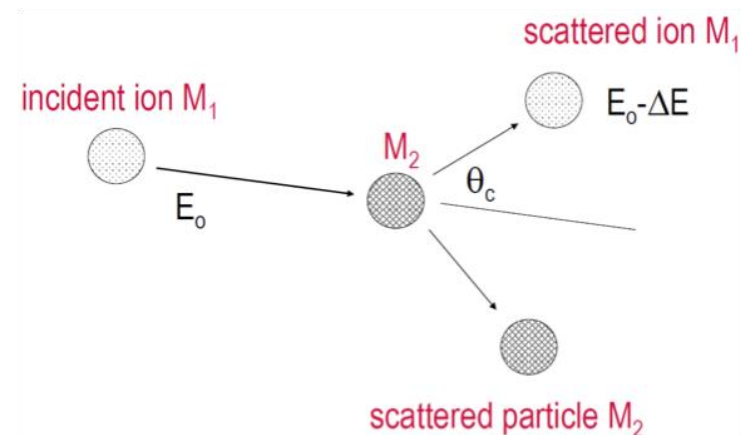
- As with CASINO, SRIM does not explicitly calculate inelastic collisions with electrons
- CASINO does not attempt to simulate damage generated by electron beams

CASINO



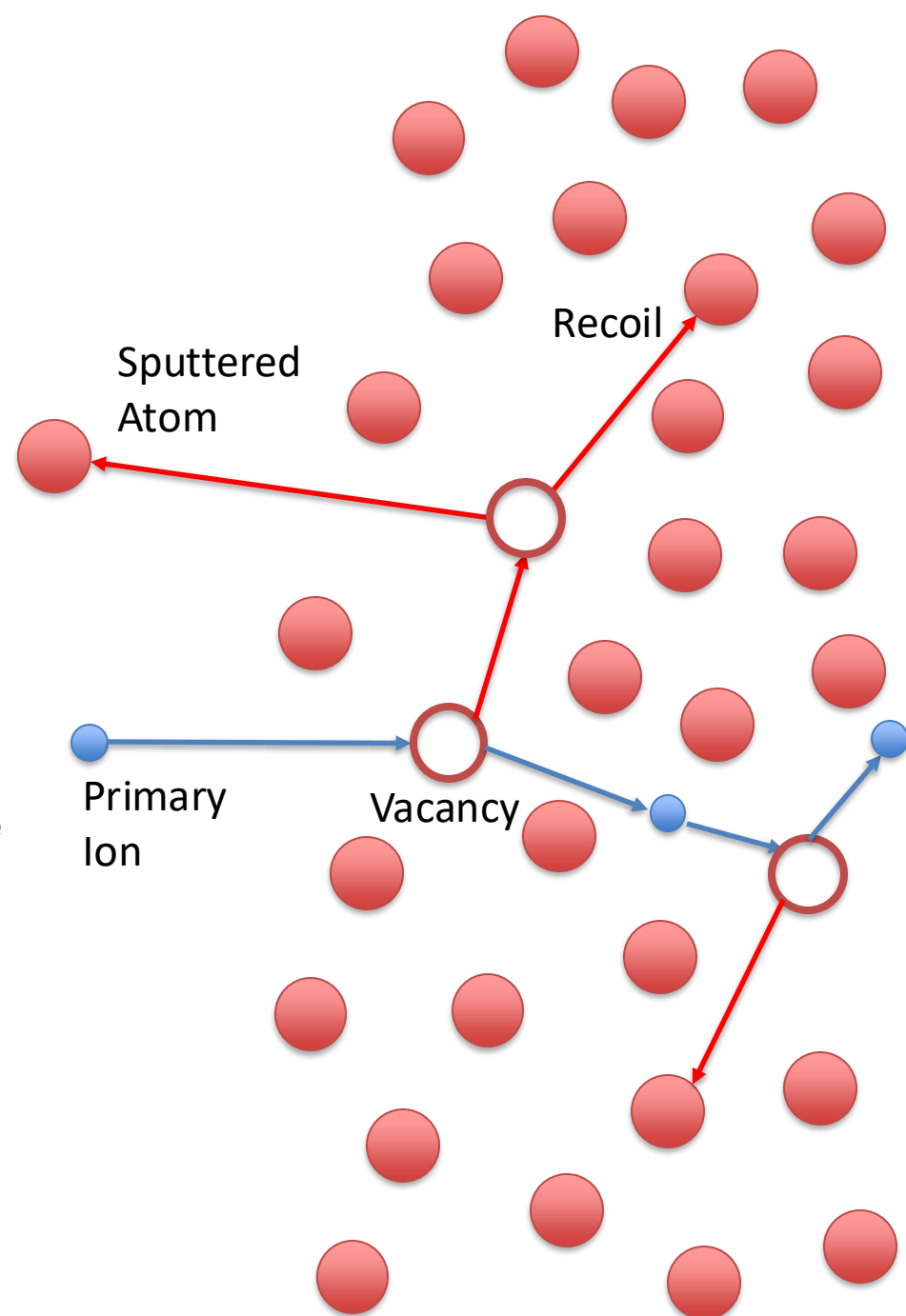
- SRIM calculates the energy transferred to a target atom in a nuclear collision, and also simulates the trajectory of the atom (called a “**recoil**”) in the sample
- SRIM has a “quick” mode in which operates similarly to CASINO

SRIM



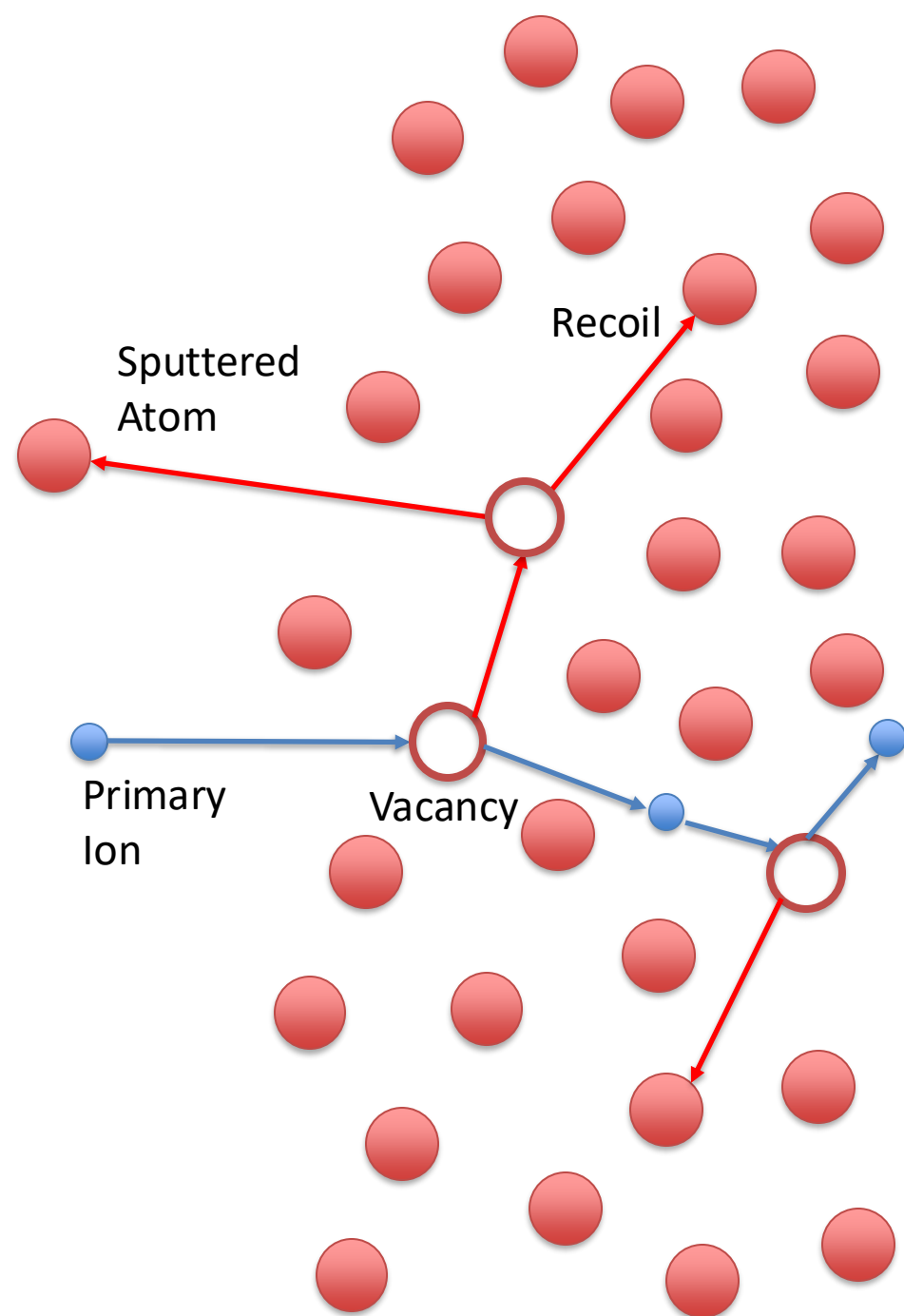
SRIM

- SRIM injects one ion into a material
- It travels some distance and loses an amount of energy according to the material's stopping power $\frac{dE}{dx}$
- At periodic intervals it encounters a target atom where the collisional energy transfer is calculated
- If the energy transferred to the target atom is sufficient, the target atom will be displaced and will form a **recoil** with energy equal to the transferred energy (ΔE) minus the displacement energy (E_d)
- The recoil then becomes a primary particle for following collisions



SRIM

- If the **recoil** makes it to the surface and has enough energy to escape, $E_k > E_{SBE}$ it is considered a **sputtered atom** and is removed from the material
- When all recoils have stopped moving, the original ion continues to its next interaction and the process repeats
- Once the original ion energy reaches zero, it has stopped and its final position is recorded (i.e., it is an **implanted atom**)
- The material then **resets** and another ion is injected

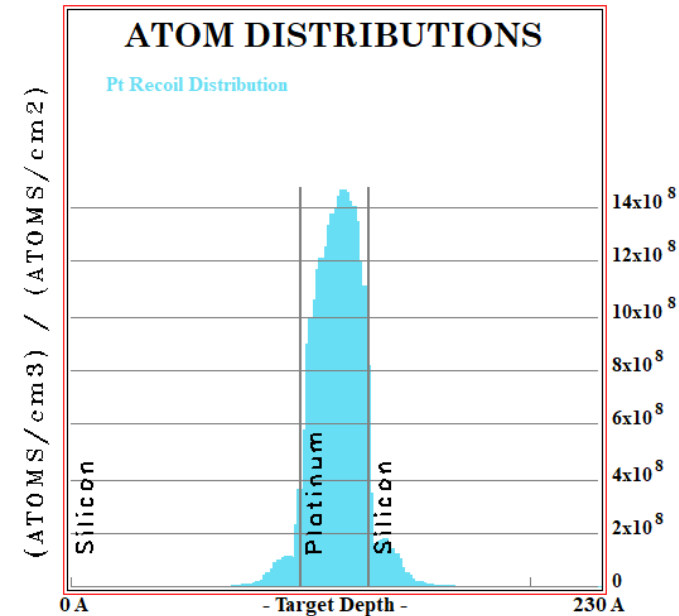
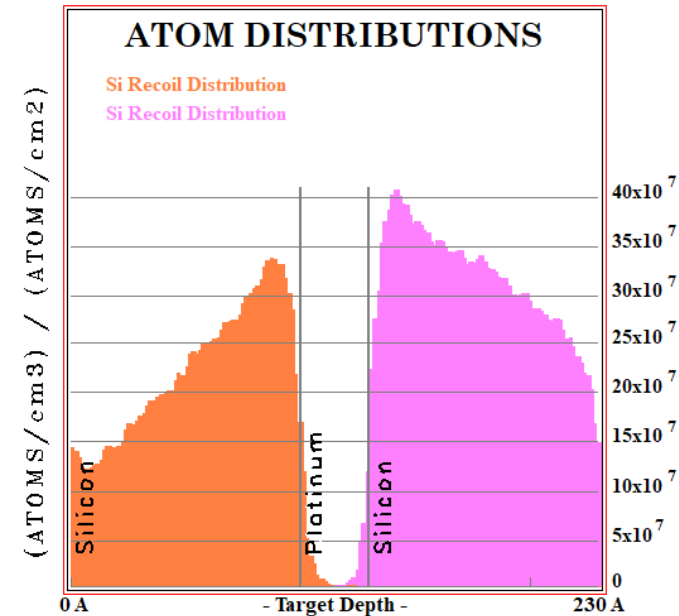


SRIM Features

SRIM can calculate:

- Material **damage** – Total displacements, vacancies, replacements.
- Ionisation – Just an approximation based on the energy loss due to electronic stopping
- **Backscattered ions** – Primary ions which escape back out of the material.
- Transmitted ions – Ions which penetrate through a thin sample.
- **Intermixing** – recoil locations in a sample consisting of 2 or more layers of thin materials.

Intermixing Example



SRIM Limitations

SRIM simulations are limited by:

- Amorphous – Channelling effects are not simulated
- Time independent – Simulations are considered to happen instantaneously (i.e., the passage of time is not simulated or recorded)
- Zero kelvin – SRIM calculates the generation of phonons but not how heating affects the sample
- Zero charge – Ions are considered to be neutral. Charge and associated electronic effects are too complex for this simulation
- Single event – Cannot calculate repeated events such as milling of material due to multiple sputtering events. Material resets after each ion

Type of calculation: Fast – No sputtering,
Full – With sputtering

Ion Species

SRIM TRIM Setup Window

TRIM (Setup Window)

Read Me ? TRIM Demo ? Restore Last TRIM Data ?

ION DATA ?

Symbol: PT, Name of Element: H, Atomic Number: 1, Mass (amu): 1.008, Energy (keV): 10, Angle of Incidence: ? 0

TARGET DATA ?

Target Layers

Add New Layer ?

Layer Name	Width	Density (g/cm3)	Compound	Corr	Gas
X Layer 1	10000	Ang 0	1		

Input Elements to Layer

Add New Element to Layer ? Compound Dictionary ?

Symbol	Name	Atomic Number	Weight (amu)	Atom Stoich or %	Damage (eV) Disp	Latt	Surf
X PT		0	1	100	20	3	2

Special Parameters

Name of Calculation: H (10) into Layer 1

Stopping Power Version: SRIM-2008

AutoSave at Ion #: 10000

Total Number of Ions: 99999

Random Number Seed: ?

Plotting Window Depths: Min 0, Max 10000

Output Disk Files

☐ Ion Ranges

☐ Backscattered Ions

☐ Transmitted Ions/Recoils

☐ Sputtered Atoms

☐ Collision Details

Special "XYZ" File" Increment (eV): 0

Buttons: Resume saved TRIM calc. ?, Save Input & Run TRIM, Clear All, Calculate Quick Range Table, Main Menu, Problem Solving, Quit

Ion energy

Layer composition (elements)

Run Simulation

Define sample layers

Layer width

Layer Density

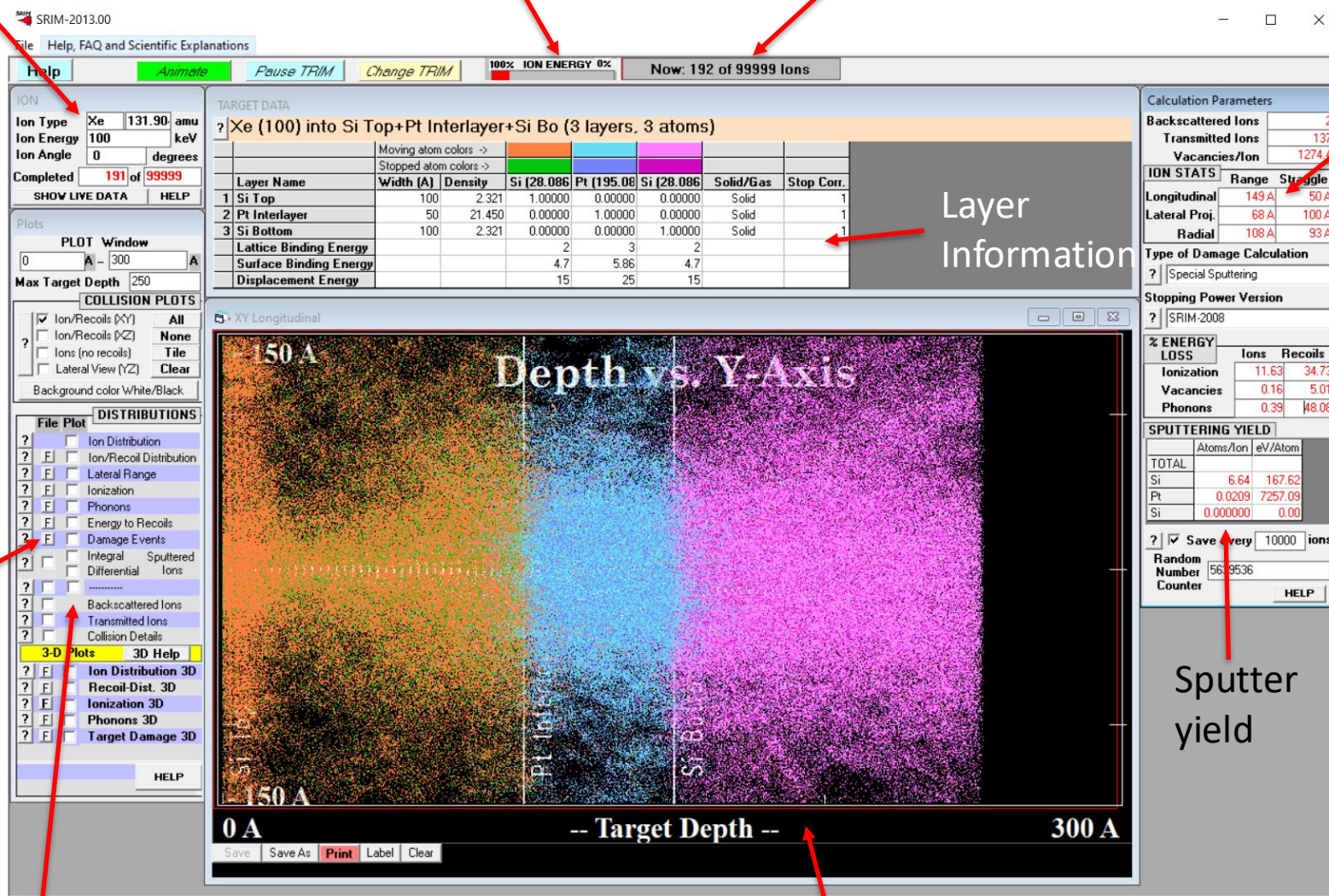
Total number of ions to simulate

Simulation
Details

Individual Ion
Progress

Simulation
Progress

Ion
Range



Layer
Information

Sputter
yield

Export
Data

Turn plots on
and off

Ion/Recoil
trajectory plot